
MA 568: Real Analysis

Course Code	MA 568
Course Name	Real Analysis
L-T-P-C	2.5-0.5-0-3
Prerequisites	IC-110: Limit, Continuity, Differentiability
Intended for	UG/MS/PhD
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **The real number system** [7 Lectures]
Sets, ordered sets, countable sets; Fields, ordered fields, least upper bounds, the real numbers, derivatives, the chain rule; Rolle's theorem, Mean Value Theorem.
- **Basic Topology** [6 Lectures]
Metric spaces, neighborhoods, open subsets, limit points, closed subsets, dense subsets; complete metric spaces, connected metric spaces, Compact sets.
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- **Sequences and Series** [6 Lectures]
Sequence, Subsequence, limits, $\limsup$ and $\liminf$; Convergence. Continuity: Continuous maps between metric spaces; Intermediate value theorem, images of compact subsets; continuity of inverse maps.
- **Convergence** [9 Lectures]
Pointwise convergence, Weierstrass criterion; continuity of uniform limits; application to power series; Spaces of functions as metric spaces, Sequence and series of functions: Uniform convergence, Uniform convergence and continuity, Equicontinuous families of functions, The Stone Weierstrass theorem.
- **Introduction to Lebesgue theory** [10 Lectures]
Set functions, Construction of Lebesgue measure, Measure spaces, Measurable functions, Simple functions, Integration.

References

1. Rudin, Walter, **Principles of Mathematical Analysis** (International Series in Pure and Applied Mathematics), 3rd Edition, McGraw-Hill, 1976.
2. Apostol, Tom M., **Mathematical Analysis**, 2nd Edition, Pearson Education, 1974.